

ex 10

English Recent Login Sign up

```

1 clear;
2 clc;
3 close all;
4
5 %% PARAMETERS
6
7 BW = [180e3 1.4e6];           % Bandwidth: NB-IoT, LTE-M (Hz)
8
9 latency = [1 0.1];           % Latency in seconds
10
11 SNR_dB = 0:5:30;             % SNR range in dB
12
13 tech = {'NB-IoT','LTE-M'};   % Technology names
14
15 %% =====
16 % 1. THROUGHPUT VS SNR
17 % =====
18
19 throughput = zeros(length(BW),length(SNR_dB));
20
21 for t = 1:length(BW)
22     for s = 1:length(SNR_dB)
23         % Convert SNR from dB to linear
24         snr_linear = 10^(SNR_dB(s)/10);
25
26         % Shannon capacity
27         C = BW(t) * log2(1 + snr_linear);
28
29         % Adjust throughput according to latency
30         throughput(t,s) = C * (1/latency(t));
31
32     end
33 end
34
35 end
36
37

```

Run Save Output Input

SNR = 25 dB : 1495.69 kbps
 SNR = 30 dB : 1794.10 kbps

LTE-M:
 SNR = 0 dB : 14000.00 kbps
 SNR = 5 dB : 28803.22 kbps
 SNR = 10 dB : 48432.04 kbps
 SNR = 15 dB : 70389.31 kbps
 SNR = 20 dB : 93214.96 kbps
 SNR = 25 dB : 116331.25 kbps
 SNR = 30 dB : 139541.17 kbps

Throughput at SNR = 20 dB

NB-IoT : 1198.48 kbps
 LTE-M : 93214.96 kbps

[Execution complete with exit code 0]

